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ORIGINAL PAPER

What is the contribution of thermal selection to the maintenance of a shell color cline in *Littorina saxatilis*?

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Abstract

The study of clines, or geographical variations of a given trait, can help understand how the interactions of genetics and local environments determine phenotypic diversity. The marine snail *Littorina saxatilis* (Olivi, 1792) exhibits a gradual change in the relative frequencies of shell color morphs across populations in the estuary system of the Rias Baixas (Galicia, NW Iberian Peninsula). A consistent pattern of distribution occurs across these four Rias, with the interior, sheltered regions dominated by a light fawn-like morph (*fulva*), and the exterior, wave-exposed locations by a darker lineated morph (*lineata*). Measurements of rock surface temperature along one of the Rias confirmed a general environmental temperature gradient during summer. The potential role of thermal adaptation driving this distribution pattern was tested by comparing shell thermal tolerance and performance between color morphs. The two color morphs (*fulva* and *lineata*) were collected from both sympatric and allopatric populations within the cline to account for the potential influence of either population or region-related traits. Laboratory experiments revealed no significant differences in the heating rate of shell temperature between color morphs in sympatric populations, although *fulva* snails showed higher recovery rates after exposure when combining sites from two Rias. As allopatric color morphs did not differ in thermal tolerance or performance, and sympatric differences were not consistent across Rias, we conclude that thermal effects represent a minor contribution to the maintenance of this color cline.

Keywords Color cline · Thermal performance · Natural selection · Littorina saxatilis

Introduction

Color polymorphism, the coexistence of multiple color morphs within a single interbreeding population, is widely observed in nature (Gray and McKinnon 2007; White and Kemp 2016). The mechanisms underpinning the maintenance of color polymorphism across spatially or temporally diverse environments have been a classic topic of study in evolutionary biology (Ford 1945; Reimchen 1979; Majerus 2009; Svensson 2017; Gefaell et al. 2023), providing key models for evolutionary theories, particularly in the context of sympatric speciation (Forsman et al. 2008; McLean and Stuart-Fox 2014). Some species display geographical color variations associated with environmental gradients such as temperature or moisture (Phifer-Rixey et al. 2008; Köhler et al. 2017; Lopez et al. 2021). These color clines can result from gradual changes in intensity (i.e. brightness or hue), or in the relative frequencies of different discrete color morphs

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across contiguous populations (Smith and Smith 2015; Gefaell et al. 2023).

In animals, color can serve multiple functions, such as signaling and communication, predator avoidance and thermoregulation (Porter and Gates 1969; Endler 1978; Caro and Allen 2017). Differences in thermoregulatory requirements and thermal tolerance among populations across a temperature gradient can lead to color clines, a phenomenon which can be explained by the thermal melanism hypothesis. This predicts that light-colored morphs of ectotherms should have an advantage in warmer conditions as they heat up more slowly and tend to maintain lower equilibrium temperatures than darker morphs, whose fast-heating rates may result in high and even detrimental body temperatures that may affect their performance and survival under thermally stressful conditions (Clusella-Trullas et al. 2007). However, the impact of coloration on the operative body temperature can be easily masked by other factors, such as predation risk, behavioral strategies, body size, or evaporative and convective heat loss (Stuart-Fox et al. 2017).

In intertidal ecosystems, especially in upper rocky shore regions, the physical environment is highly complex and dynamic, with extreme fluctuations in temperature, desiccation stress, and salinity occurring on a daily basis governed by the rise and fall of tides (Raffaelli and Hawkins 1999; Little et al. 2009). During the low tide period, organisms are emersed and exposed to solar radiation and wind, which drive their body temperatures (Helmuth 2002), reaching extreme levels on shores where wave splash is limited (Denny et al. 2009). With the diverse thermal environment across rocky shores, these represent ideal systems to study the thermal effects of coloration and its relationship with physiological, morphological, or behavioral adaptations of ectotherms (Etter 1988; Judge et al. 2018).

The intertidal rough periwinkle, *Littorina saxatilis* Olivi (1792), exhibits an extensive degree of phenotypic variation (in terms of shell size, shape and color), and has been widely used as a model in evolutionary studies (Rolán-Alvarez et al. 2015; Johannesson 2016; Gefaell et al. 2023). This marine snail is one of the most common species on rocky shores of the North Atlantic, from wave-exposed coasts to wave-sheltered habitats such as estuaries or salt marshes, feeding on microalgae and organic detritus on the rock surface (Reid 1996). The ovoviviparous reproductive system and limited dispersal capability of this species contribute to striking local adaptations (Rolán-Alvarez et al. 2015). While these adaptations are mainly associated with shell size and shape differences, shell color variation across populations has also been documented at both local and regional geographic scales (Gefaell et al. 2021, 2024). Notably, a 21-year analysis by Johannesson and Butlin (2017) provided evidence of selection acting on *L. saxatilis* color morphs, as supported by work on similar littorinid species (Reimchen 1979; Phifer-Rixey et al. 2008; Gefaell et al. 2025).

The Rias Baixas in Galicia (NW Iberian Peninsula), comprising four consecutive coastal inlets formed by the immersion of four different fluvial valleys during the Holocene transgression (Pagés-Valcarlos 2000; Arce-Chamorro et al. 2022), are a particularly interesting region to study shell color polymorphism in *L. saxatilis*. Each inlet (ria) is characterized by a consistent pattern showing an overall shift from a light fawn-like morph (*fulva*) in the interior (near the river mouth), wave-sheltered environment, to a darker lineated morph (*lineata*) in the exterior, more exposed coastal environment (Sacchi 1979; Reid 1996; Gefaell et al. 2024b). While *fulva* is the dominant morph in most sites in the interior region (frequencies exceeding 50%), the exterior sites show remarkably higher frequencies of *lineata*, with even monomorphism (100% *lineata*) found in some of them (Gefaell et al. 2024b). In contrast, great color diversity is observed in populations from intermediate regions, where these two morphs coexist, along with other color morphs (white, yellow, orange, brown or black; Sacchi 1979; Reid 1996; Gefaell et al. 2024b). Furthermore, this color morph distribution or color cline has remained relatively stable over the past forty years at least in the more extensively studied Ria de Vigo (Sacchi 1979; Gefaell et al. 2024b), suggesting the presence of a selective force stabilizing the cline.

Certain shell color polymorphisms and clines within the genus *Littorina* have been previously proposed to be maintained by thermal adaptation (Phifer-Rixey et al. 2008). Environmental temperature may, therefore, be a contributing factor shaping the shell color cline in the Ria de Vigo, assuming a temperature gradient exists along the Rias, with inner regions reaching the highest temperatures consequent to decreased wave exposure. Following the thermal melanism hypothesis (Clusella-Trullas et al. 2007), the low frequency of *lineata* in the interior regions of the Rias could be attributed to a greater influence of solar radiation in these wave-sheltered environments. In littorinid snails, shell temperature is a good proxy for body temperature (Lathlean et al. 2016; Seuront and Ng 2016), thus differences in shell heating rates due to shell coloration may be directly correlated to the performance and fitness of *L. saxatilis* individuals in areas with different temperature regimes.

To verify our hypothesis proposing temperature as a contributing (non-exclusive) factor to the maintenance of this color cline in the Rias Baixas in Galicia, rock surface temperature was monitored along the color cline using autonomous temperature loggers. Shell heating rates and thermal tolerance in air of the dominant color morphs (*fulva* and *lineata*) were assessed under laboratory simulated low tide scenarios. We studied shell heating rates under natural sunlight

outdoors mimicking near-field conditions, and under heat lamps in the laboratory to test the effects of exposure to more extreme temperatures. To assess thermal tolerance, we measured both lethal and sublethal (foot attachment or recovery after thermal exposure) responses of snails to different temperatures, either in air in a temperature-controlled water bath or under heat lamps. Specifically, we compared morphs collected from sympatric populations (i.e. coexisting snails which are exposed to similar local environments and so any differences in their thermal responses can be primarily attributed to their shell color differences), as well as allopatric populations (i.e. comparisons of color morphs from different populations over a large geographic scale) to understand the role of thermal adaptation as an evolutionary mechanism maintaining color polymorphism in natural populations of *L. saxatilis* along the rias of the NW Iberian Peninsula.

Materials and methods

Animal collection, maintenance and experimental design

Adult specimens of *L. saxatilis* were collected during low tide at five sites, covering three of the Rias Baixas (Galicia, NW Iberian Peninsula; Fig. 1a). The two dominant color morphs, *fulva* (with a plain light shell) and *lineata* (with a lineated dark shell), were collected (see Fig. 1b), from both sympatric (where both coexist at intermediate frequencies) and allopatric (where they do not coexist and are respectively found at high frequencies) populations. Three sympatric populations across three rias were assessed to ensure that any observed thermal responses are not the result of localized processes inherent to any specific ria. In sympatric populations, coexisting morphs are exposed to similar environmental conditions and, assuming similar behavior, any differences in their thermal responses could be attributed to the shell color differences among other shell traits. Despite their different position along the cline (with some populations being further inside the ria in the Ria de Vigo; Fig. 1a), the sympatric populations we sampled show analogous local conditions in terms of wave exposure, being partially sheltered due to the orientation relative to the open sea and proximity to an island, and microhabitat heterogeneity, which might also explain the coexistence of both morphs (*fulva* and *lineata*) at similar frequencies (see Vigo et al. 2025). In contrast, comparing morphs across allopatric populations allows the evaluation of how different local environments influence these snails across a broader geographical scale. Findings on sympatric versus allopatric populations, therefore, allow us to ascribe the potential differences in thermal responses between morphs to either color, by affecting morphs from a sympatric population, or to other traits that could be correlated with color when affecting exclusively monomorphic allopatric samples. Adult snails were brought to the Toralla Marine Science Station (ECIMAT) and kept in aquaria for 1–4 d prior to the experimental measurements. Sampling and acclimation details are included in **Supplementary Methods 1**. To monitor the natural thermal environment along the cline, we used autonomous temperature loggers (EnvLogger T7.3, ElectricBlue, Vairão, Portugal) that recorded the rock surface temperature (T_r) every 30 min from August 2023 to July 2024 at 12 sites of the Ria de Vigo (Fig. 1a). Each logger was placed in the intertidal zone, at those sites where individuals of *L. saxatilis* were sampled by Gefaell et al. (2024b) to ensure a more accurate representation of the thermal conditions experienced by the snails (see **Supplementary Methods 2**).

Thermal tolerance and body heating rates in a sympatric population

To assess thermal tolerance and heating rates of the two color morphs, snails of both morphs were haphazardly collected from a sympatric population at Punta Cubillo (Ria de Vigo). To measure their thermal tolerance, snails were heated up constantly in the laboratory until reaching one of the seven experimental temperatures (T_{exp}; from 41 to 47 °C at 1 °C intervals according to preliminary trials), then maintained at this designated temperature for a duration of either 1 min (i.e. acute exposure) or 1 h (i.e. chronic exposure) and subsequently examined for pedal responses. T_{exp} were randomized in sequence to minimize the potential effects of any temporal cycles.

For each T_{exp}, ten snails of each color morph were transferred to individual empty vials (30 mL) with the operculum facing down, allowing foot attachment and normal movement in air within the vial. Lids were loosely attached to prevent the snails from crawling out while allowing air exchange during the experiments. Vials were immersed (with the lids above the water line to maintain aerial conditions inside the vials) in a programmable water bath (Grant TXF200, Grant Instruments, Cambridge, UK) held at room temperature (23 °C) and ramped up at a constant rate (~1 °C 10 min^{-1}, which is close to the maximum heating rates observed from previous rock temperature data in similar locations; see also Dwane et al. 2023) until reaching the T_{exp}. To monitor the change in snail body temperature (T_b) during heating, thermocouples were inserted through the operculum of one snail from each color morph (these individuals were not used in tolerance measurements).

Once the T_{exp} was reached, the heating was stopped and the bath maintained at the designated temperature. Five

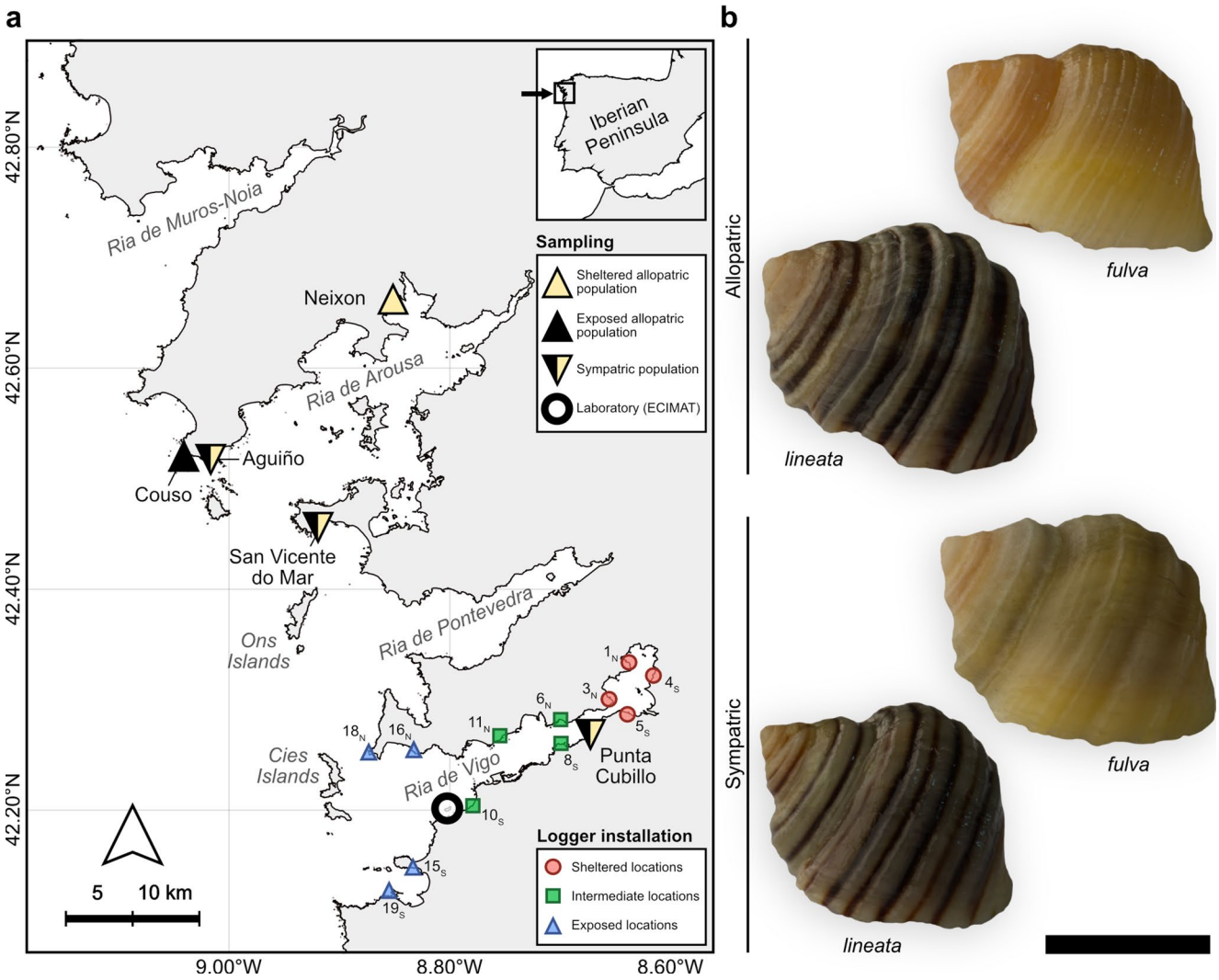


Fig. 1 Sampling and rock surface temperature monitoring sites along the cline and color morphs used in the laboratory experiments. (**a**) Rias Baixas and geographic location of the sampling sites within each Ria (triangles). The location of Toralla Marine Science Station (ECIMAT) is also shown (open circle), as well as the sites where the temperature loggers were installed, covering sheltered (red circles), intermediate (green squares) and wave-exposed (blue triangles) locations according to Gefaell et al. (2024b; see their Table 1 for coordinates and more information about each location). (**b**) Visual appearance of the color morphs from both allopatric and sympatric populations. Scale bar: 5 mm. All the images were taken in RAW format with a Nikon D500 digital camera attached to a 105 mm f/2.8 Sigma macro lens and calibrated using a white balance reference card to standardize and compare shell coloration (Stevens et al. 2007). Maps were created using shapefiles from Instituto Geográfico Nacional (for the Rias Baixas; BDLJE CC-BY 4.0, ign.es) and Natural Earth (for the Iberian Peninsula; naturalearthdata.com) datasets in QGIS (qgis.org), then modified in Affinity Designer 2 (affinity.serif.com)

individuals of each morph were removed either immediately within 1 min of exposure or after 1 h, and then transferred to dishes of seawater, allowing for recovery overnight at room temperature (20–22 °C). Thermal performance and tolerance were assessed using foot attachment and response to a pedal stimulus as sublethal and lethal responses respectively ($\Sigma n = 5$ individuals × 7 T_{exp} × 2 exposure durations × 2 color morphs = 140 snails). Finally, snails were sexed in vivo by observing the presence of a penis under a stereo microscope.

To compare the heating rates of the two color morphs under near-field conditions, body temperature of snails exposed outdoors under natural sunlight were monitored. Snails sampled from Punta Cubillo (Ria de Vigo) were haphazardly chosen from the aquaria and two morphs of similar size (≤ 2 mm difference) were paired. A small hole (< 1 mm diameter) was drilled into the dorsal apex of the snails to allow insertion of a thermocouple (type K, Lutron, Taiwan) to measure T_b (Hamby 1975; Seuront et al. 2018). Each pair of snails were superglued with operculum facing down (to mimic natural on-shore attachment via mucus) to a rock tile within 5 cm of each other. The position of the pairs was alternated to minimize spatial effects. The rock tiles were placed outdoors at ECIMAT in sunny, wind protected

locations to simulate an emersion period and T_b was monitored every 5 min for ~4 h each day. New pairs of snails were set up each day to ensure temperature was monitored for living snails with minimal water loss. In total, four days of monitoring were performed (Σn = 2 color morphs × 4 rock tiles × 4 d = 32 snails).

Shell heating and recovery in the laboratory

Snails were sampled from sympatric populations in two of the Rias Baixas (Aguiño, Ria de Arousa and San Vicente do Mar, Ria de Pontevedra), as well as from allopatric populations (*lineata* from Couso and *fulva* from Neixon) in the Ria de Arousa. Thermal exposure for shell heating measurements was performed under heat lamps at four different T_{exp} (35, 40, 45, 50; ± 1.0 °C) for all populations and morphs (Σn = 14 individuals × 2 color morphs × 4 T_{exp} × (2 sympatric populations + 1 pair of allopatric populations) = 336 snails).

Each snail was stimulated to retreat inside the shell, and any excess water was blotted dried with absorbent paper to minimize measurement errors due to water reflectance (Lathlean and Seuront 2014). The snails were placed individually with their operculum facing down in the center of a 2 L glass beaker under a terrarium heat lamp (Intense Basking Spot 150 W, Exo-Terra, Rolf C. Hagen Inc., Montreal, Canada; ~30 cm between the snail and the lightbulb) controlled by a thermostat (ITC-308, INKBIRD Tech. C. L., Shenzhen, China), maintaining each glass beaker at one of the four T_{exp} (35, 40, 45, 50; ± 1.0 °C; with the thermostat probe maintained at the same height in all the beakers) before and during the experiment (**Supplementary Fig. 1a**, **Supplementary Methods 3**). After 10 min, shell temperature (T_{sh}) was measured twice using infrared thermography (IRT; Testo 883, Testo SE & Co. KGaA, Baden-Württemberg, Germany; Caddy-Retalic et al. 2011; Chapperon and Seuront 2011; Seuront et al. 2018). All thermal images were analyzed using testo IRSoft Software (v. 5.0; testo.com) and the maximum temperature point of the shell surface was extracted for each image and averaged (n = 14) as the T_{sh} (**Supplementary Fig. 1b**, **Supplementary Methods 3**). Additionally, the same measurements were carried out on empty shells (obtained by removing the soft tissues after boiling) from sympatric populations, using a sample of Σn = 5 individuals × 2 color morphs × 2 sympatric populations = 20 snails. Each empty shell was exposed to all four T_{exp} for measurements (i.e. 80 observations).

Recovery from thermal exposure in live snails was evaluated immediately following IRT imaging. Snails were allowed to recover in seawater at room temperature (18–20 °C) for 30 min, after which any responses including movement, foot attachment, or response to pedal stimulus (poking the foot with fine forceps; see Sandison 1967; McMahon and Payne 1980) were recorded as successful recovery. The rest of the snails were classified as unrecovered, indicating heat coma or even mortality in some snails.

Statistical analyses

Temperature data along the cline

Rock temperature data along Ria de Vigo (T_r) were summarized into hourly averages per logger and then divided into four seasons (summer: June-August; fall: September-November; winter: December-February; spring: March-May). These groups, as they are sequential due to Northern Hemisphere seasonality, account for the statistical dependance of temperature data across months given their similar climatic conditions (Shimura et al. 2013), as well as the ecological relevance within the rocky shore because of the seasonality of the tides (Stickle et al. 2017; Wada et al. 2017). Furthermore, the UTC standard was adopted to avoid local time changes. Season dependence was assessed by comparing the mean T_r per logger (location, n = 12) between seasons using one-way ANOVA and corresponding Tukey post-hoc tests. Then, for each season, a linear regression model was performed with the mean T_r as the dependent variable and an ecological estimate of wave exposure as the predictor. This estimate was provided by a previous principal component analysis (PCA) comparing the overall ecological conditions across these locations, including variables such as distance from the river mouth and presence or absence of organisms (other gastropods, barnacles and algae) typically reflecting wave exposure (see Gefaell 2024) to illustrate the thermal gradient associated with the different *L. saxatilis* populations. As 86% of the variance of this ecological PC1 value is explained by the distance from the river mouth, this PC1 value can be extracted as an ecological proxy for the degree of wave exposure, with the smallest values corresponding to sheltered locations and the highest to the most exposed locations (see Gefaell 2024). The same linear regression model was conducted using a randomized subsample (n = 200 observations per location) during low tide periods (observations under the average sea level, retrieved from Puertos del Estado tide gauge for Vigo; portus.puertos.es) in summer.

Thermal tolerance and body heating rates in a sympatric population

To compare the response to thermal exposure, percentage foot attachment (i.e. a sublethal response) and percentage survival (a lethal response) were tested separately using binary logistic regression (BLR) models, with the effects

of color morph (fixed, two levels: *lineata*, *fulva*), exposure duration (fixed, two levels: acute, chronic), sex (fixed, two levels: male, female) and their interactions tested. Experimental temperature was not included as a predictor in these models as the data suffered from quasi-complete separation (i.e. temperature yielded an almost perfect prediction of the response, such as an outcome of 0 or 100% regardless of other variables). However, those experimental temperatures at which the dependent variable differed between shell color morphs were assessed by Fisher's exact tests. In addition, the temperatures at which 50% of individuals lost foot attachment (AT_{50}) or died (LT_{50}) were estimated using BLR models with T_{exp} as a predictor. This LT_{50} value determined from chronic exposure was then used to evaluate the extent of thermal stress snails would experience along the Ria de Vigo by assessing the duration of each location exceeding this temperature threshold. Given the unequal number of measurements among locations (July 2023 was not covered in two of the locations), the percentage duration (estimated as the proportion of hours above LT_{50} to total hours within each season) was used as the response variable in a linear regression model with PC1 as the predictor for each season. This was also assessed for the entire year.

To compare body heating rates across color morphs, the T_b difference between the morphs (computed as *lineata* - *fulva*; positive values indicate hotter *lineata*) was calculated for each pair of snails. These differences were tested against zero using Welch's t-tests for each day and each time point ($n = 4$ pairs of snails; with rock tiles, the experimental units as replicates).

Shell heating and recovery in the laboratory

ANOVA was performed on the data collected from sympatric populations from both the Ria de Arousa and the Ria de Pontevedra, using color morph (fixed, two levels: *lineata*, *fulva*), T_{exp} (fixed, four levels: 35, 40, 45 and 50 °C), ria (fixed, two levels: Arousa, Pontevedra) and their interactions as the predictors and T_{sh} (natural logarithm to achieve homoscedasticity) as the dependent variable. Population type (sympatric or allopatric population) was assessed within the Ria de Arousa, running the same analysis but with population type (fixed, two levels: sympatric, allopatric) instead of ria as a factor. Lastly, recovery after the experiment was studied for all data (including all sympatric and allopatric populations) using a dichotomous recovery variable (0 = unrecovered, 1 = recovered) as the dependent variable using a binary logistic regression and the same predictors as above. Fisher's exact tests for each T_{exp} were used to assess possible differences between shell colors under every scenario.

For the additional sample of empty shells, a log-linear regression model was carried out using color morph, T_{exp}, ria and their interactions from sympatric populations as predictors and T_{sh} (natural logarithm) as the dependent variable. ANOVA was used to further study the effect of shell color, T_{exp} and their interaction (predictors) on T_{sh} (natural logarithm) as the dependent variable. Population type (sympatric or allopatric population) was also tested by ANOVA within the Ria de Arousa using population type (fixed, two levels: sympatric, allopatric) instead of ria.

All the models were fitted using 'lm', 'glm', 'aov' and 'fisher.test' functions within the "stats" package in R Statistical Software (v. 4.4.3; r-project.org).

Results

Temperature data along the cline

The change in T_r over a year in each location was variable. Some locations showed particularly narrow temperature ranges (1_N and 6_N) and, overall, the sheltered region reached higher T_r and experienced the greatest seasonal differences (e.g. between summer and winter) (**Supplementary Fig. 3**). Average environmental temperatures were significantly higher in summer (ANOVA, $F(3, 44) = 284.56$, $P < 0.05$, see **Supplementary Fig. 4**). When only considering this season, the sheltered region reached the highest temperatures overall ($P < 0.05$; **Supplementary Table 2**; Fig. 2) although this is not consistent during low tide periods, when T_r showed no significant association with wave exposure (Linear regression, adjusted $R^2 = -0.02$, $F(1, 10) = 0.73$, $P = 0.41$; Fig. 2), as well as the proportion of hours above LT_{50} (44 °C according to results of the thermal tolerance experiment in the Ria de Vigo; Linear regression, adjusted $R^2 = 0.06$, $F(1, 10) = 1.74$, $P = 0.22$).

Thermal tolerance and body heating rates in a sympatric population

Generally, snails attained normal locomotory functions (100% foot attachment) and survivorship (100% survival) upon exposure to temperatures around 41 °C and showed a decrease of such functionality and survivorship as temperatures and exposure duration increased (Fig. 3). The interaction between color morph, sex, and their respective effects on foot attachment or survival were found to be non-significant overall, while the exposure duration (i.e. acute versus chronic) did affect both responses (**Supplementary Tables 3 and 4**): the temperature under which 50% the snails lost foot attachment (AT_{50}) or died (LT_{50}) was ~2.5 °C lower under chronic exposure (Fig. 3). When comparing the performance

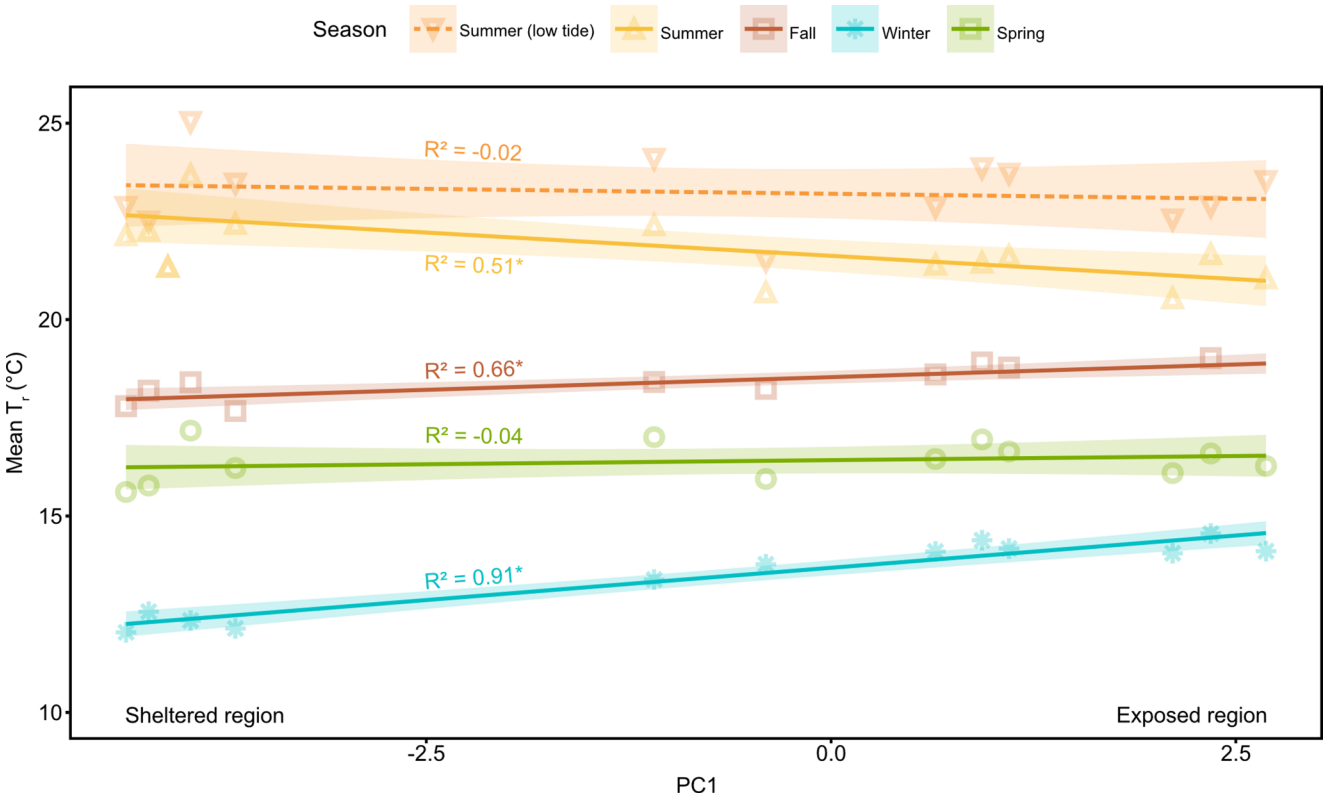


Fig. 2 Scatter plots of mean T_r by season (different shape and color) against ecological PC1 value representing distance along the Ria as an indicator of wave exposure along the Ria de Vigo (with lower PC1 values towards the more sheltered region). Each linear regression is shown with its adjusted R^2 value, followed by an asterisk (*) if the trend is significant (summary in **Supplementary Table 2**). 95% confidence intervals are also shown. The plots were generated in R (r-project.org), then edited in style using Affinity Designer 2 (affinity.serif.com)

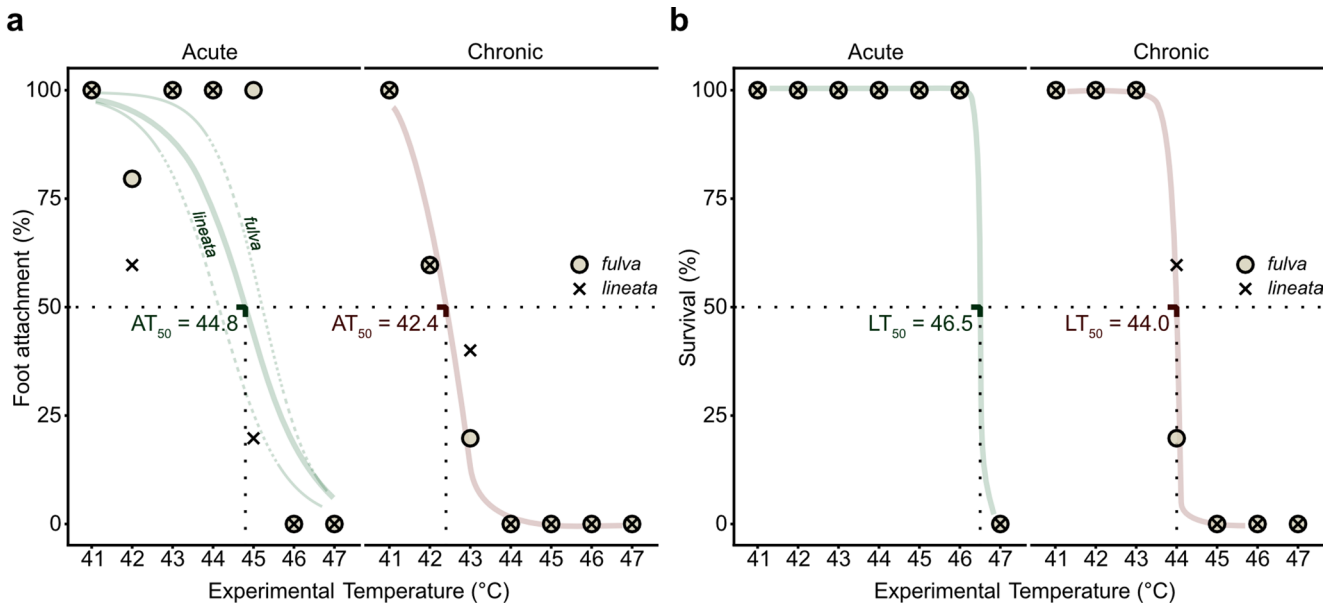


Fig. 3 Thermal response of snails from a sympatric population in the Ria de Vigo to different temperatures and durations of exposure. (**a**) Percentage of foot attachment (thermal performance) and (**b**) survival (thermal tolerance) after either acute or chronic exposure to each experimental temperature for both color morphs (*fulva* and *lineata*). The continuous lines indicate AT_{50} and LT_{50} estimations based on linear regressions using experimental temperature as predictor for foot attachment and survival, respectively. Color morphs were pooled given their minimal contribution (with the exception of AT_{50} for foot attachment under acute exposure, where dashed lines were drawn separately for each color morph). The plots were generated in R (r-project.org), then edited in style using Affinity Designer 2 (affinity.serif.com)

between color morphs across the studied temperatures, the only observed difference occurred after an acute exposure at 45 °C (Fisher's exact test, $P=0.048$; Fig. 3a), which is reflected in the higher AT_{50} of *fulva* (45.3 °C) compared to *lineata* (44.3 °C). In all other cases, the differences between color morphs were either nonexistent or minimal (Fisher's exact test, $P>0.05$; Fig. 3a). This pattern was also observed for thermal tolerance, particularly under acute exposure, showing both morphs a 100% survival rate from 41 to 46 °C, and then experiencing a full mortality at 47 °C (Fig. 3b). For chronic exposure, this 100% survival rate ended at 43 °C, and no alive snails were found at temperatures above 44 °C (Fig. 3b). At 44 °C, although survival differed between color morphs, the effect of shell coloration was again negligible (Fisher's exact test, $P>0.05$).

T_b was not significantly affected by shell coloration when exposed to natural sunlight. In the four days of experiments, the mean T_b differences between pairs of color morphs were not significant except for one day, where *lineata* were almost a degree hotter than *fulva* for ~1.5 h (**Supplementary Fig. 5**). Overall, no significant differences in body heating rates of the two morphs were found (Welch's t-test, $P>0.05$).

Shell heating and recovery in the laboratory

No differences in shell heating between color morphs (Fig. 4) or its interaction with T_{exp} were found in sympatric populations, although there was a significant effect of the interaction between T_{exp} and ria (**Supplementary Table 5**). For snails from the Ria de Arousa, the thermal effect of shell coloration was only noticeable between populations, depending on whether these were sympatric or allopatric (**Supplementary Table 6**).

In terms of recovery from heat exposure, *fulva* snails recovered at higher rates in comparison to *lineata* snails overall (Fig. 5), although this difference was not statistically significant among rias and population types (Binary logistic regressions; **Supplementary Tables 7 and 8**). When focusing on each combination, the differences between color morphs were more evident in sympatric populations at 35 and 40 °C. Although non-significant for each ria separately (Fisher's exact test, $P=0.098$ at 35 °C and $P=0.385$ at 40 °C in the Ria the Arousa; $P=0.098$ at 35 °C and $P=0.054$ at 40 °C in the Ria the Pontevedra), the higher tolerance of *fulva* snails was enhanced when pooling sympatric populations from both rias at these T_{exp} (Fisher's exact test, $P=0.004$ at 35 °C and $P=0.023$ at 40 °C). Snails from

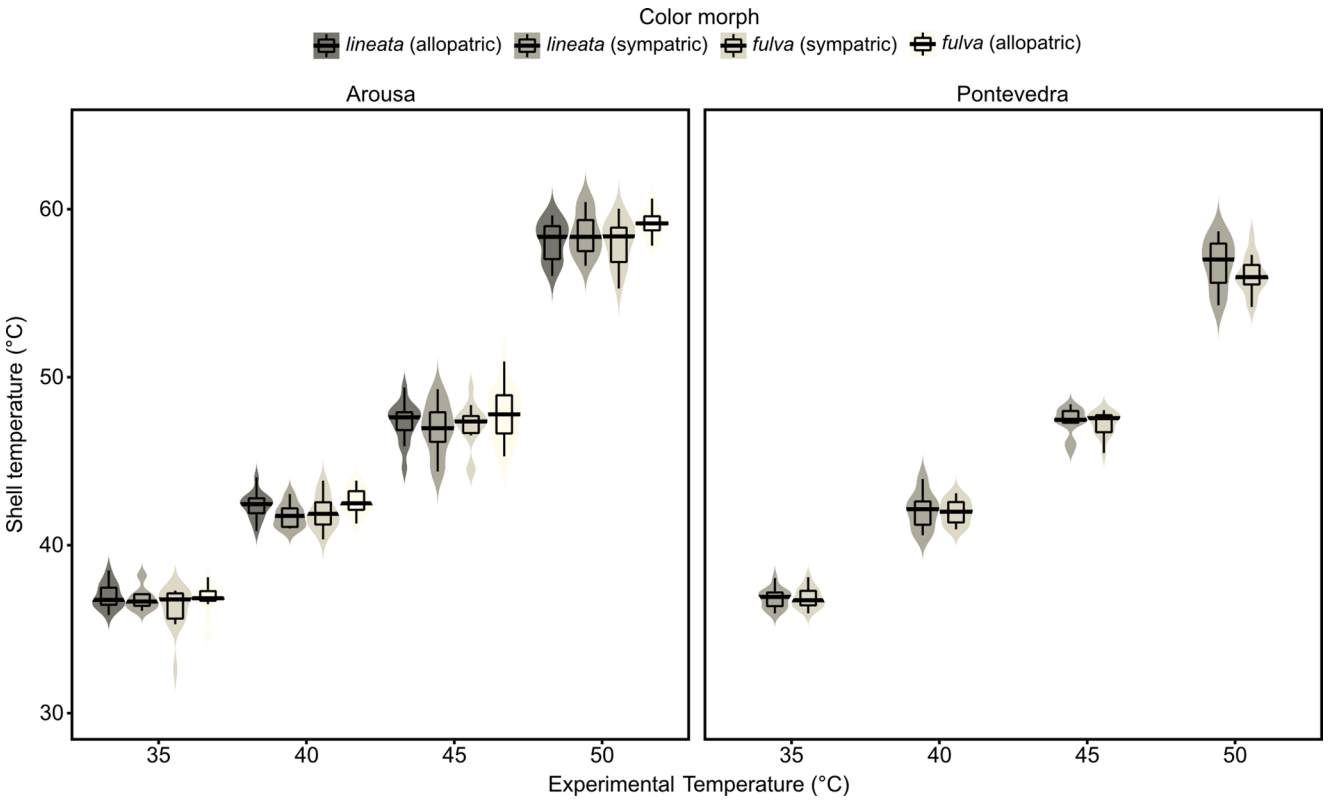


Fig. 4 Shell temperature after a 10-min exposure of both color morphs for each population type (allopatric or sympatric) and ria to different experimental temperatures. Boxplots show the median for T_{sh} (bold line), the interquartile range and the 5th and 95th percentiles (whiskers). Violin plots represent the probability density of the data, with the width of the shaded area indicating the proportion of observations with that value. The plots were generated in R (r-project.org), then edited in style using Affinity Designer 2 (affinity.serif.com)

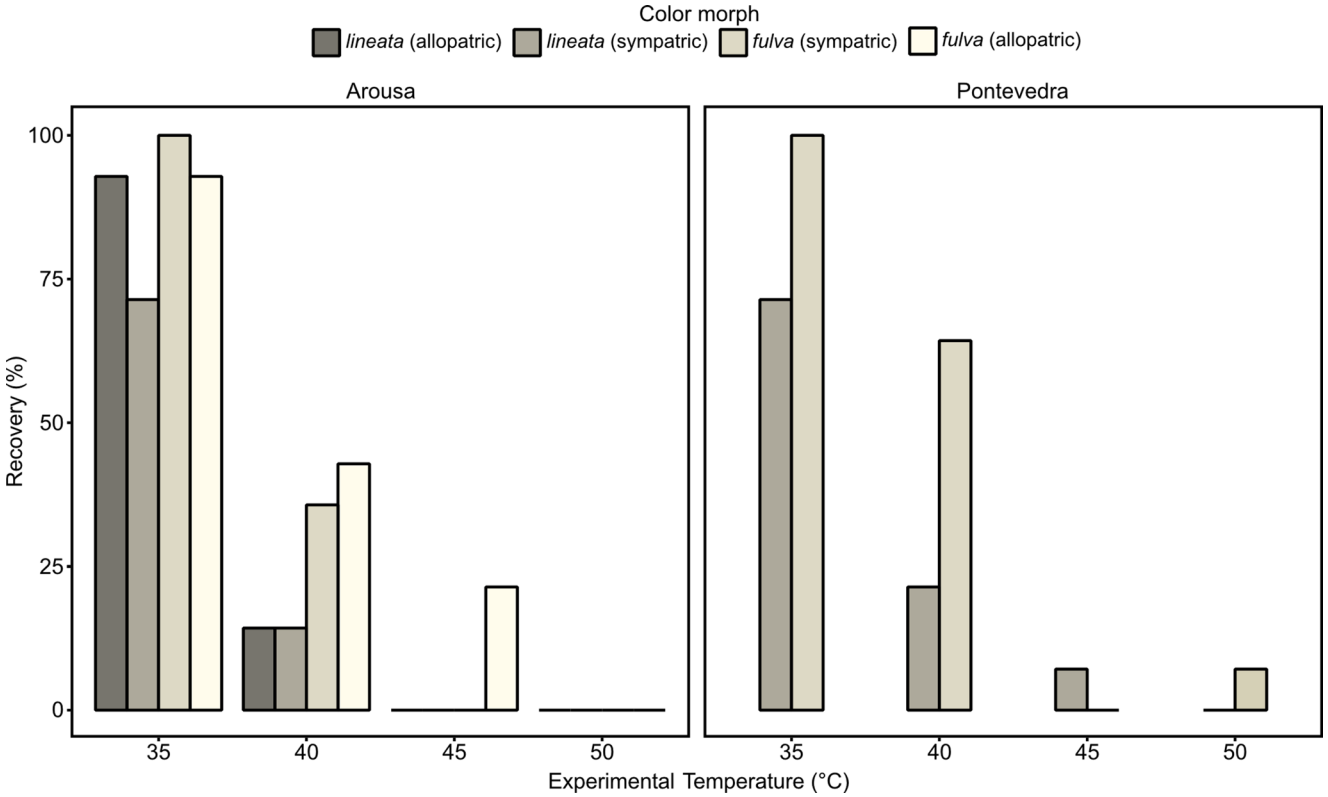


Fig. 5 Recovery rate after thermal exposure of both color morphs, population types (allopatric or sympatric) and rias. Rates refer to the number of recovered snails divided by the total number of snails exposed to heat lamps for 30 min after the shell temperature measurements. The plots were generated in R (r-project.org), then edited in style using Affinity Designer 2 (affinity.serif.com)

allopatric populations from the Ria de Arousa achieved high recovery rates (92.9%) at 35 °C irrespective of their color morph, but *fulva* was, in line with sympatric populations, more tolerant at 40 °C although these differences were not significant (Fisher's exact test, $P=0.209$ at 40 °C; Fig. 5). Most snails were, however, unable to recover under higher temperatures (Fig. 5). In the Ria de Arousa, 21.9% of allopatric *fulva* snails recovered at 45 °C, while *lineata* snails did not show any activity following the recovery period. Nonetheless, the effect of shell coloration on recovery was also not significant in this case (Fisher's exact test, $P=0.222$).

Additionally, shell coloration and its interaction with T_{exp} had no effect on T_{sh} in empty shells (**Supplementary Tables 9 and 10**). However, within the Ria de Arousa (**Supplementary Table 11**), snails from allopatric populations showed significantly different temperatures when compared to sympatric populations (ANOVA test, $P<0.05$), although there was no interaction with color morphs (ANOVA test, $P=0.522$).

Discussion

Our aim was to assess the potential role of thermal stress in maintaining the observed shell color cline of *L. saxatilis* in the Rias Baixas. We identified an overall environmental temperature gradient along the Ria de Vigo during summer, with the inner, sheltered region having higher average temperatures than the outer, wave-exposed region. Despite this, when comparing the two most representative color morphs in sympatric populations from this and two other Rias, we found no overall consistent differences in their thermal tolerance, performance or body heating rates. Interestingly, the *fulva* morph achieved higher recovery rates following heat exposure at high but not extreme temperatures, particularly in sympatric populations. Given the fact that the observed differential responses were not more variable at the extremes of the cline (i.e. in allopatric populations), we suggest the contribution from thermal stress to the maintenance of the overall color cline pattern is insufficient to be considered as a major selective factor by itself.

The direction of the thermal gradient measured along the Ria de Vigo differed across seasons. In summer the mean T_r was higher in the more wave-sheltered locations, generally towards the inner part of the ria, when solar radiation

is expected to be the greatest. This mean temperature is determined based on the variations throughout complete tidal cycles, encompassing surface water temperature during high tide and substratum temperature when emersed during low tide. When exposed during low tide, substratum temperature is influenced by wind, solar radiation, as well as ambient air temperature (Marshall et al. 2010). Although air temperature was not directly measured in this study, it has been reported to be highly correlated with surface water temperature in the Rias Baixas, and the maxima are lower in the outermost regions (Nogueira et al. 1997; Alvarez et al. 2005). As such, our T_r measurements represent a good proxy of environmental temperature experienced by the snails living in close proximity, which is known to correlate well with shell and body temperatures (assuming limited behavioral thermoregulation; Marshall et al. 2010; Ng et al. 2017; Judge et al. 2018). In the context of our study, to assess whether shell coloration contributes to differential thermal response under heat stress, the emersion periods, especially in summer, are particularly relevant. Although T_r sometimes surpassed 50 °C during summer in half of the locations, which exceeds the acute thermal tolerance of these snails (46.5 °C) and may predict an impact on snail survival and fitness with direct sunlight exposure, mean T_r and cumulative time exposure above 44 °C (i.e. chronic thermal tolerance of these snails) did not differ between the extremes of the Ria under emersion. These results must be interpreted with caution, since rock temperature data were only recorded along the Ria de Vigo and do not coincide with our sampling sites. It is worth noting, however, the thermal similarities among the Rias Baixas, as reported by Alvarez et al. (2005).

Laboratory experiments did not detect a major color morph difference in their thermal responses, which suggests there is no correlative relationship with the color cline and the observed temperature gradient in the Rias Baixas. Thermal performance and tolerance (sublethal and lethal responses), after acute and chronic exposure to heat were similar in the dark (*lineata*) and light (*fulva*) morphs. The notable exception was *fulva* snails showing significantly higher performance following acute exposure at 45 °C (100% foot attachment, in contrast with < 25% of the *lineata* morph), which suggests future studies should focus on the temperature range between 44 and 46 °C, as no snails were able to maintain their foot attachment under higher temperatures irrespective of their shell color. The LT_{50} values obtained (44–46.5 °C with 1 min to 1 h of aerial exposure) are slightly higher when compared to previous studies of the same species, which can be attributed to the different populations and ecotypes sampled, as well as measurement methods (previous studies estimated lethal limits when submersed; Evans 1948; Sandison 1967; Clarke et al. 2000). The AT_{50} values (42.4–44.8 °C with 1 min to 1 h of aerial exposure) are generally < 2 °C below the lethal temperatures, supporting the idea that the collapse of physiological function occurs below the critical thermal limit (Dwane et al. 2023). On the shore, the inability to recover from thermal exposure and loss of locomotory function can be considered ecological mortality, given that the relaxation of the foot muscle would make the snail more vulnerable to external threats such as predation (Truebano et al. 2018; Dwane et al. 2021). However, the differences between lethal temperatures and those resulting in loss of locomotory function observed here, are lower than those reported in previous studies (~ 10 °C when submerged; Evans 1948; Clarke et al. 2000; ~6 °C when emersed; Sandison 1967). This discrepancy may be partially due to the different methods used in scoring "coma" (loss of locomotory function), which vary from pedal response to prickling (i.e. a simple neurological reflex) to more coordinated foot attachment (which better represents sustained performance under heat stress, as used in this study). It is worth noting that LT_{50} and AT_{50} values determined under laboratory conditions may not reflect responses under more stochastic field conditions, calling for caution when extrapolating laboratory results to natural populations (Rezende et al. 2014; Drake et al. 2017; Ng et al. 2017).

Significant, but relatively subtle differences were observed between sympatric and allopatric populations. Within the Ria de Arousa, the color morphs showed differences in shell heating rates when compared to allopatric populations, where, in contrast to the thermal melanism hypothesis (Clusella-Trullas et al. 2007), *fulva* reached higher temperatures than *lineata* when exposed at 50 °C. A significant shell color-population type (allopatric or sympatric) interaction could indicate a relationship between temperature and the distribution of color morphs. However, given that heating rates only differed between color morphs from each extreme of the cline (i.e. allopatric monomorphic populations), but not within sympatric populations, any contribution of temperature must occur in combination with other correlated traits. Color genes have been located on chromosomal inversions in *L. saxatilis* (Westram et al. 2018; Koch et al. 2021), which may be associated with thermoregulation genes and selected only under extreme conditions. The environmental relevance of this observation is, however, questionable as differences were only observed at 50 °C, well above the acute LT_{50} (46.5 °C) of the snails. Interestingly, differences between color morphs were detected in recovery rates upon thermal exposure under heat lamps when data of sympatric populations from two rias (Ria de Arousa and the Ria de Pontevedra) were pooled, where *fulva* showed a higher recovery rate than *lineata* after exposure to 35 and 40 °C. Exposure above 40 °C did not

lead to significant differences between color morphs, but under such extreme temperatures, most snails were in coma or dead (acute AT_{50} = 42.4 °C). Despite a few cases of significant color morph differences (only in some measurements of sympatric populations), the overall similarity of thermal responses in allopatric color morphs suggests limited contribution of environmental temperature in maintaining the color cline across the rias.

Whether thermal adaptation contributes to the maintenance of color clines appears to be species specific. *Theba pisana*, a highly polymorphic land snail that has been widely studied regarding shell coloration, shows similar color morphs as those found within this color cline of *L. saxatilis*. Consistent with our results, some studies of this land snail also found no differences between a pale (*fulva*-like) and a darker (pigmented pattern similar to our *lineata* morph) color morph in terms of shell thermal equilibrium using either empty shells (Scheil et al. 2012) or living snails (Knigge et al. 2017). Knigge and co-workers proposed that the darker morph would instead heat up and cool down significantly faster than the light morph. In contrast, other studies reported darker morphs to be consistently hotter in terms of shell and body temperatures when compared to paler morphs, suggesting that snails with darker shells would be more susceptible to experiencing elevated thermal stress (Köhler et al. 2013, 2021). In marine snails, a latitudinal color cline in the Gulf of Maine has been reported to be maintained by temperature/desiccation in *Littorina obtusata* (Schmidt et al. 2007; Phifer-Rixey et al. 2008). Both empty shell temperature and mortality of *L. obtusata* significantly differed between color morphs, with dark shells getting warmer and the corresponding snails less likely to survive than the light morph, supporting the thermal melanism hypothesis. According to theoretical biophysical models, both heating rates and equilibrium temperatures would be predicted to be higher for darker individuals under high incident radiation, low ambient temperature and no wind conditions (Clusella-Trullas et al. 2007). Seuront et al. (2018) mimicked these conditions in a field experiment where they compared four color morphs of Galician *L. saxatilis* populations under both direct sunlight and shade. Although the darker morph warmed up significantly faster, they found no differences in body temperature. This could be because the shell temperature differences as a result of radiation may be negated by conductive or convective heat exchange, which can be prominent in small animals. For example, Miller and Denny (2011) recorded significant temperature differences between black and white shells in five littorinid species, but such differences diminished with increasing contact and thus conductive flux with the rock substratum. Sun exposure also affected the heating rate since the snails warmed up twice as fast under direct exposure, as observed in other intertidal gastropods (Franklin et al. 2022). Indeed, the body temperature of littorinids has been reported to be unaffected by shell coloration or other traits, such as size or shape. Instead, behavioral strategies such as shell lifting, towering and habitat selection are more likely to exert a more pronounced effect on body temperature (Miller and Denny 2011; Marshall et al. 2013; Chapperon et al. 2017; Ng et al. 2017). Interestingly, some studies report that the association of littorinids with barnacles may mitigate prolonged thermal stress during low tide, especially in summer (Cartwright and Williams 2012, 2014; Moisez et al. 2020), while others show that *L. saxatilis* tends to reach heat coma later, i.e. at higher temperatures, in wave-exposed intertidal locations (Clarke et al. 2000). This phenomenon might be observed in the outer region of the color cline in the Rias Baixas where the dark morph (*lineata*) tends to form monomorphic populations (Gefaell et al. 2024a, 2024b) and which is also characterized by dense barnacle coverage (*Semibalanus balanoides* and *Chthamalus* spp.; Gefaell et al. 2025).

In conclusion, we found no systematic differences in thermal performance and tolerance among *L. saxatilis* color morphs. Consequently, the thermal gradient along the intertidal shores of the Rias Baixas in summer does not appear to be the main driver of the distribution of shell coloration in this species. Given the lack of evidence for thermal adaptation as a primary driver for the observed color cline, we suggest that this cline is more likely the result of a combination of other physiological and ecological factors, such as differential predation (Johannesson and Butlin 2017; Gefaell et al. 2025), or selection acting on other genes within the same inversion as color (Koch et al. 2021).

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Author contributions GAW, MT and ERA conceived the original idea. RV, JGe, JGa and ERA installed temperature loggers and collected the corresponding data. RV, SLL and JGe collected snail samples. RV and SLL conducted laboratory experiments, the corresponding data analysis and prepared Tables and Figures. RV analyzed temperature logger data and wrote the original draft. All authors contributed to the manuscript and approved the final version for submission.

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Data availability All data and analysis codes are openly available at https://github.com/ramonvigoq/ThermalSelection_ColorCline.

Declarations

Competing interests The authors declare no competing or financial interests.

Ethical approval All snails were collected from locations with sufficient density to avoid compromising the population viability. No special permissions were required for these experiments as *L. saxatilis* is not considered a vulnerable species. However, the wellbeing of all the individuals involved in the experiments was ensured according to the ASAB guidelines (ASAB Ethical Committee/ABS Animal Care Committee 2023). Surviving snails from the Ria de Vigo were returned to their habitat after the experiment. Snails from the Ria de Arousa and Pontevedra were euthanized by freezing at -20 °C immediately following the recovery period.

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